LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need and types of Static converting equipment - rectifiers, transformers, transformer rectifier units, rotary inverters, static inverters and automatic change over system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need and types of Static converting equipment - rectifiers, transformers, transformer rectifier units, rotary inverters, static inverters and automatic change over system. Practical on inverter, transformer & rectifier.
5. Specific Objectives	13.1.2.1 Knowledge of Static converting equipment, Bus bars, cable terminations, Switches.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on Inverter, Transformer & Rectifier: Inspect internal components, verify cable terminations, identify switch positions, and test continuity. Safety: Wear PPE, ensure power isolation, verify zero voltage before touching terminals. Supervision: Instructor monitors safety compliance and component identification accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a rectifier in an aircraft power system?
(b) Define the role of a transformer in static converting equipment.
(c) Why is an inverter required in modern avionics power distribution?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need for Static Converting Equipment • Rectifiers and Transformer Rectifier Units • Transformer construction and types • Cable terminations and bus bar connections	L & Ds	PPT, IB	15 min
(b)	• Rotary Inverters and Static Inverters • Automatic Change Over System operation • Comparison of Inverter types • Safety precautions for high voltage equipment	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of Inverter • Inspection of Transformer components • Verification of Rectifier connections • Identification of Switches and terminations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a rotary inverter and a static inverter based on their mechanical components.
(b) Describe the sequence of operation for an automatic change over system during a power failure.
(c) Identify three critical safety precautions required when handling transformer rectifier units.

12. Home Assignment and Follow Up	Prepare a brief note on the specific requirements for cable terminations on bus bars as discussed in LP 02, highlighting the importance of secure connections for avionics reliability.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Bus bars: - Bus bar systems. Block/schematic explanation of bus bar systems, split busbar systems, concept of emergency busbar. Practical on busbar interconnection system.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min